

ResNet · DenseNet

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Introduction

Network

- ☐ ResNet
 - Main idea
 - Architecture
- ☐ DenseNet
 - Main idea
 - Architecture
- ☐ Comparison

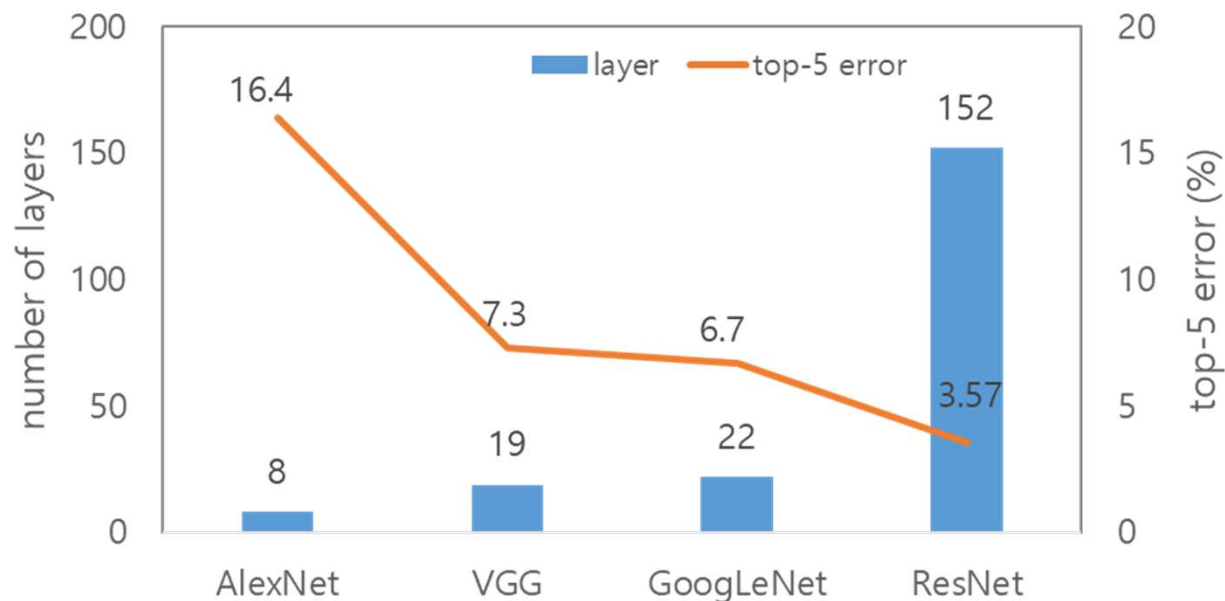
Experiment

- ☐ Setup
- ☐ Result

Conclusion

- ☐ Future Work

Neural Network Performance

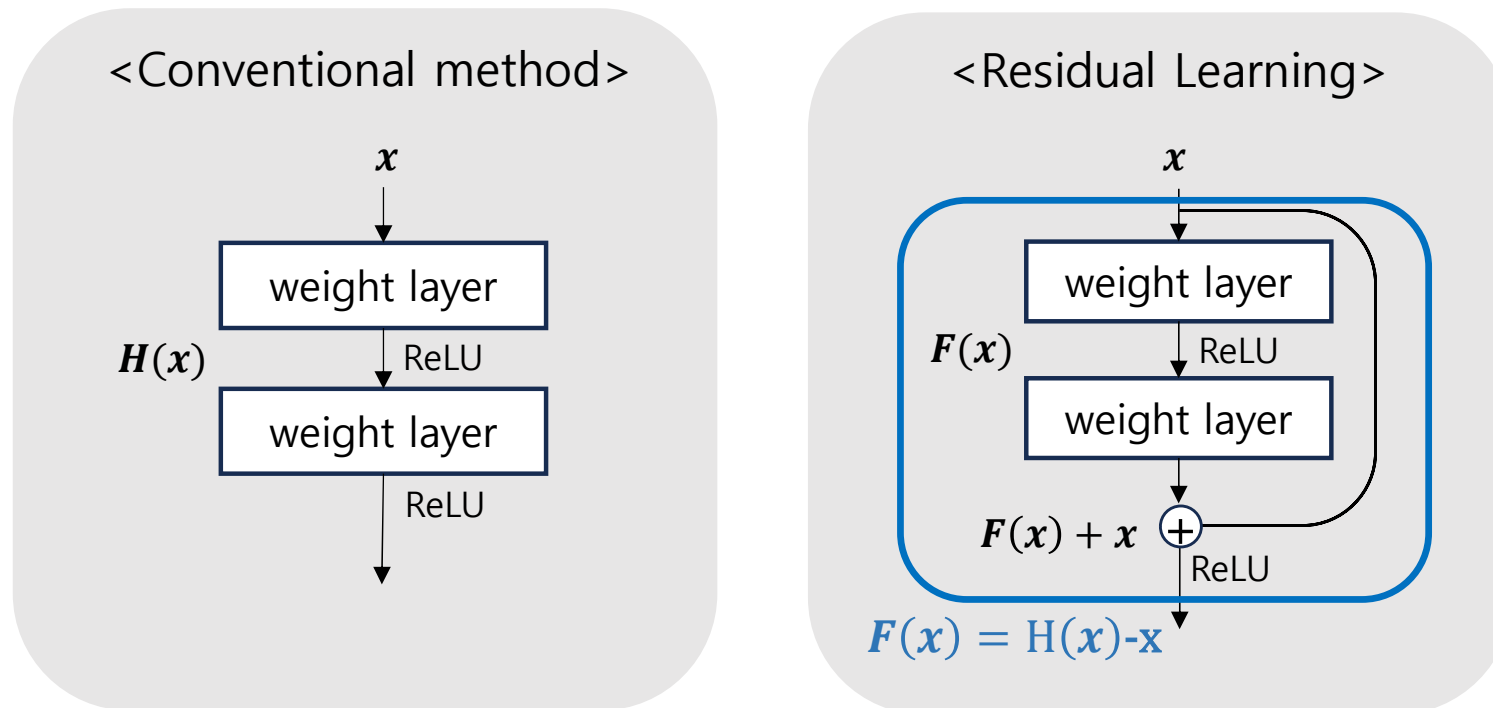


Shortpath
idea



■ Main Idea

1. Residual Learning

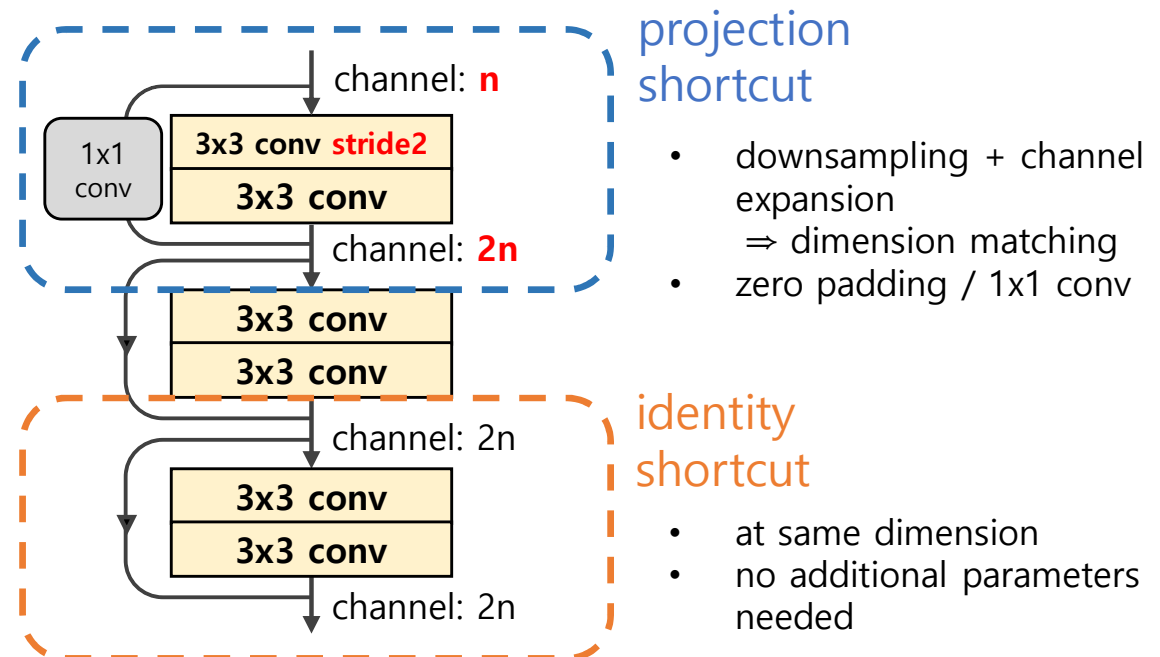
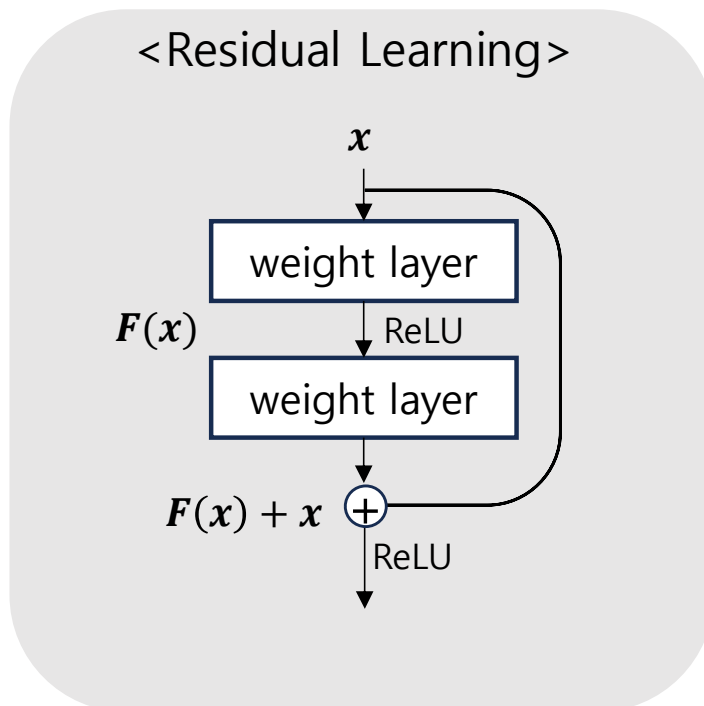


- If optimum is identity function, It is easier to get the difference.
- Information and gradient loss less at forward/ backward propagation

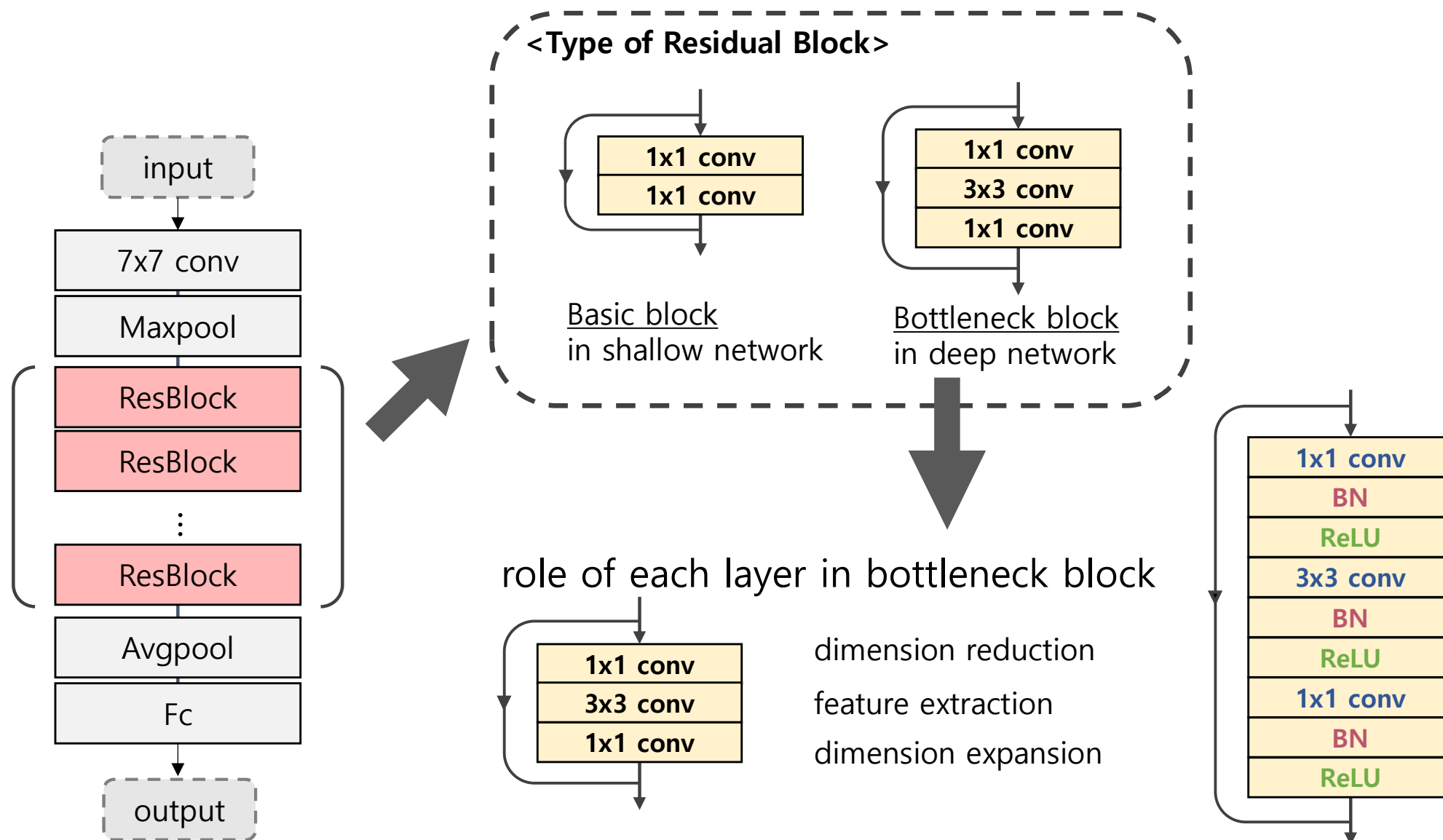
■ Main Idea

2. Shortcut Connection

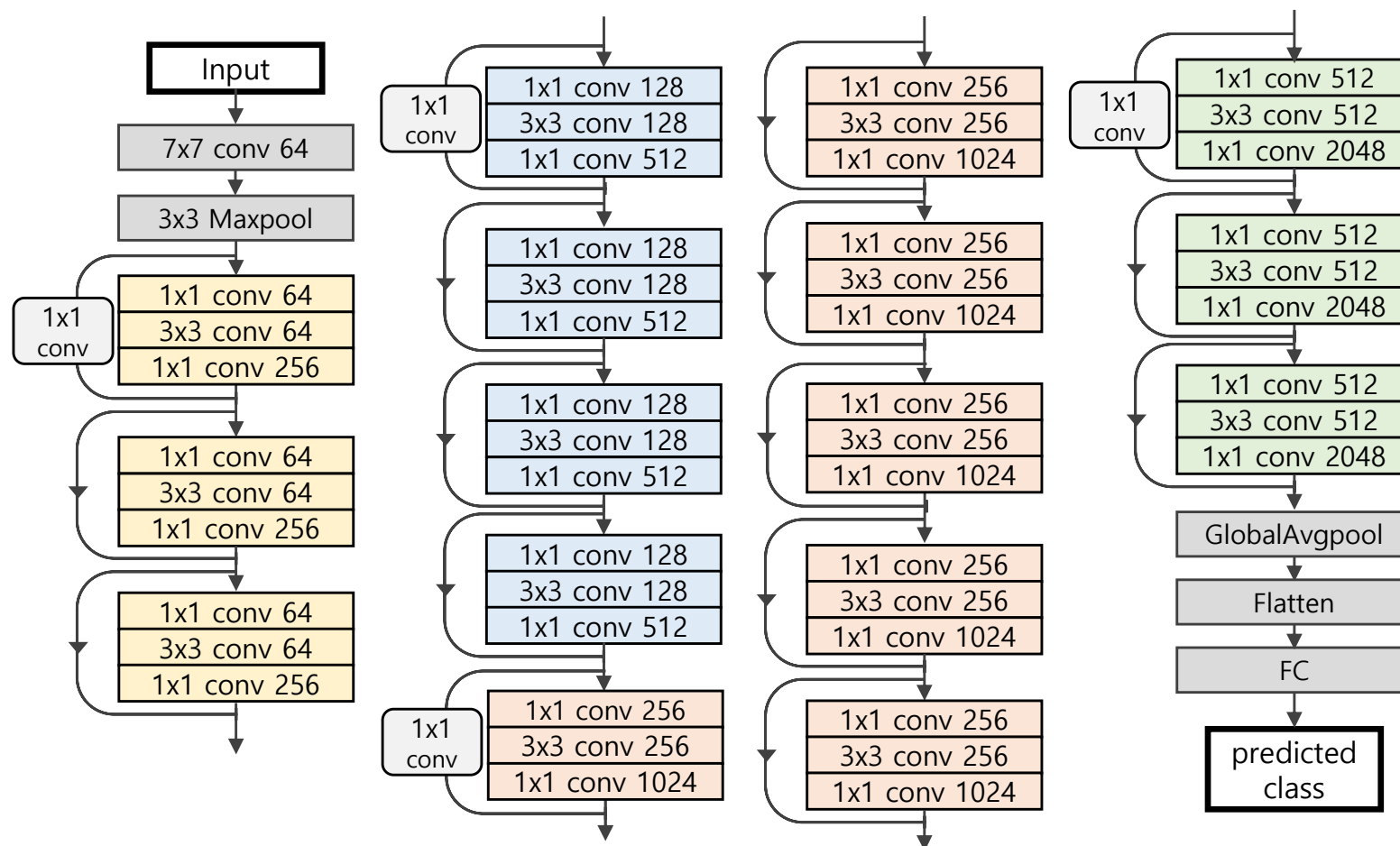
- Identity Shortcut vs Projection Shortcut



Architecture

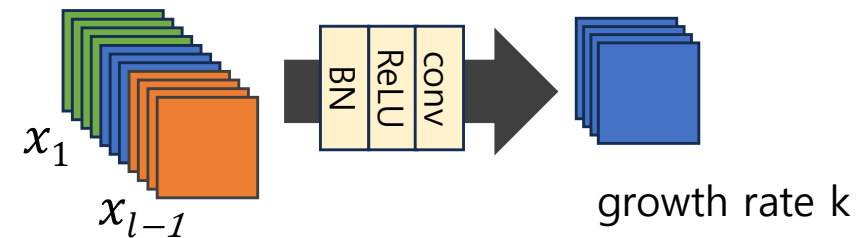
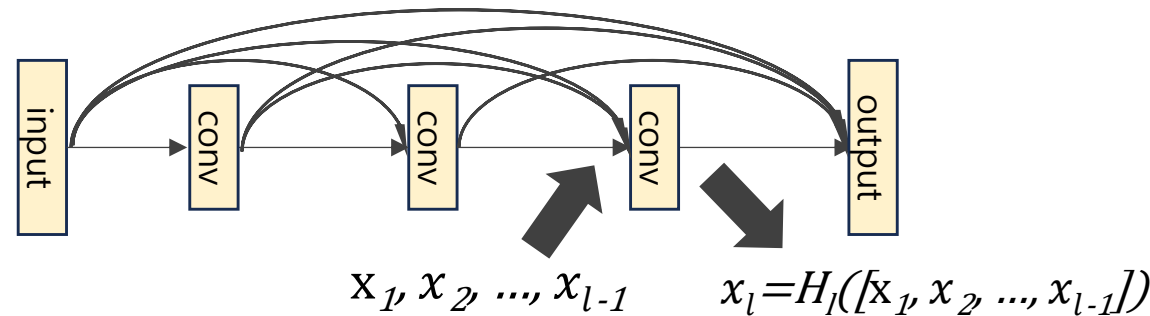


Architecture of ResNet50



■ Main Idea

Dense Connectivity



Advantages

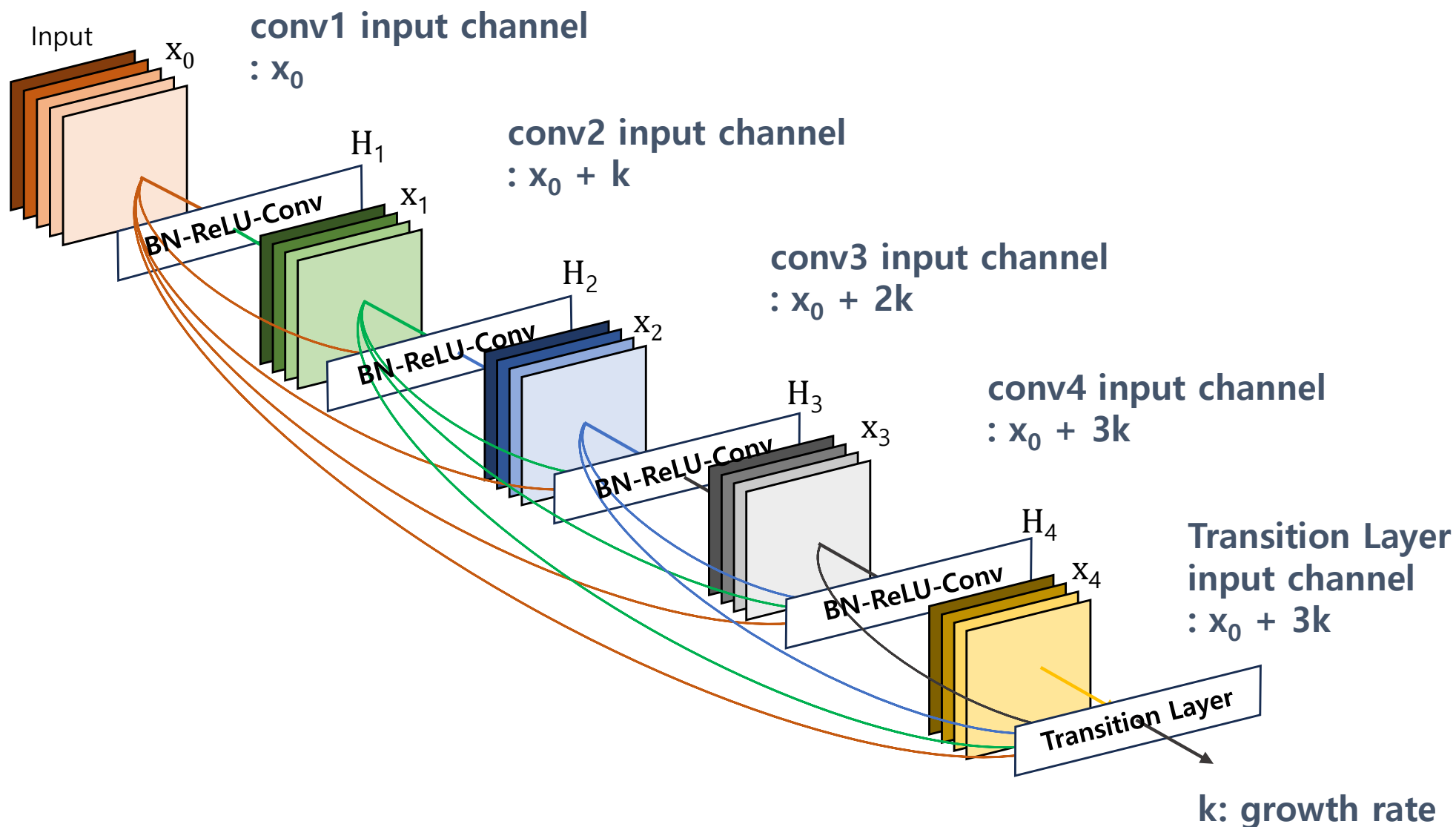
1. parameter efficiency

- why? no need to relearn redundant feature-maps
- $\Rightarrow$ can prevent overfitting
- $\Rightarrow$ can make deeper network

2. improved flow of information and gradients

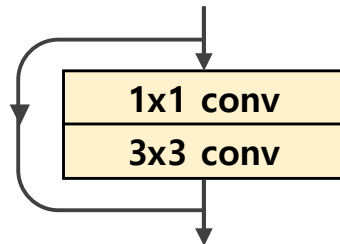
- $\Rightarrow$ easy to train

Architecture



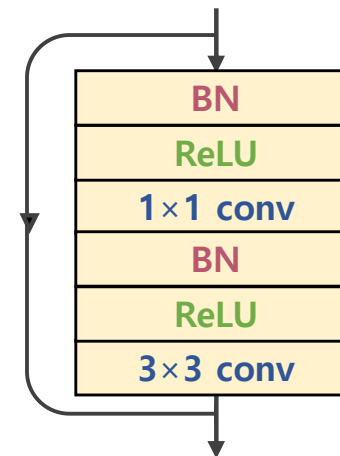
■ Architecture

1. Bottleneck layer

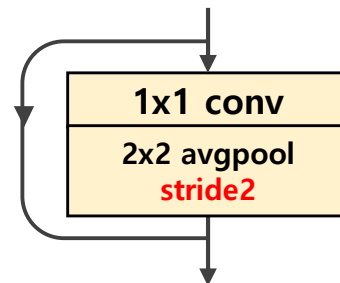


Concatenation acts like an expansion
⇒ 1×1 conv is not used at the end

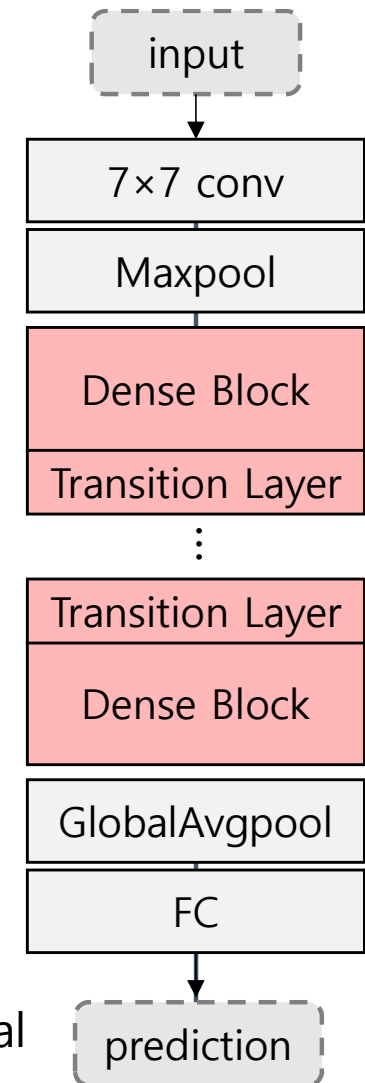
* preactivation



2. Transition layer

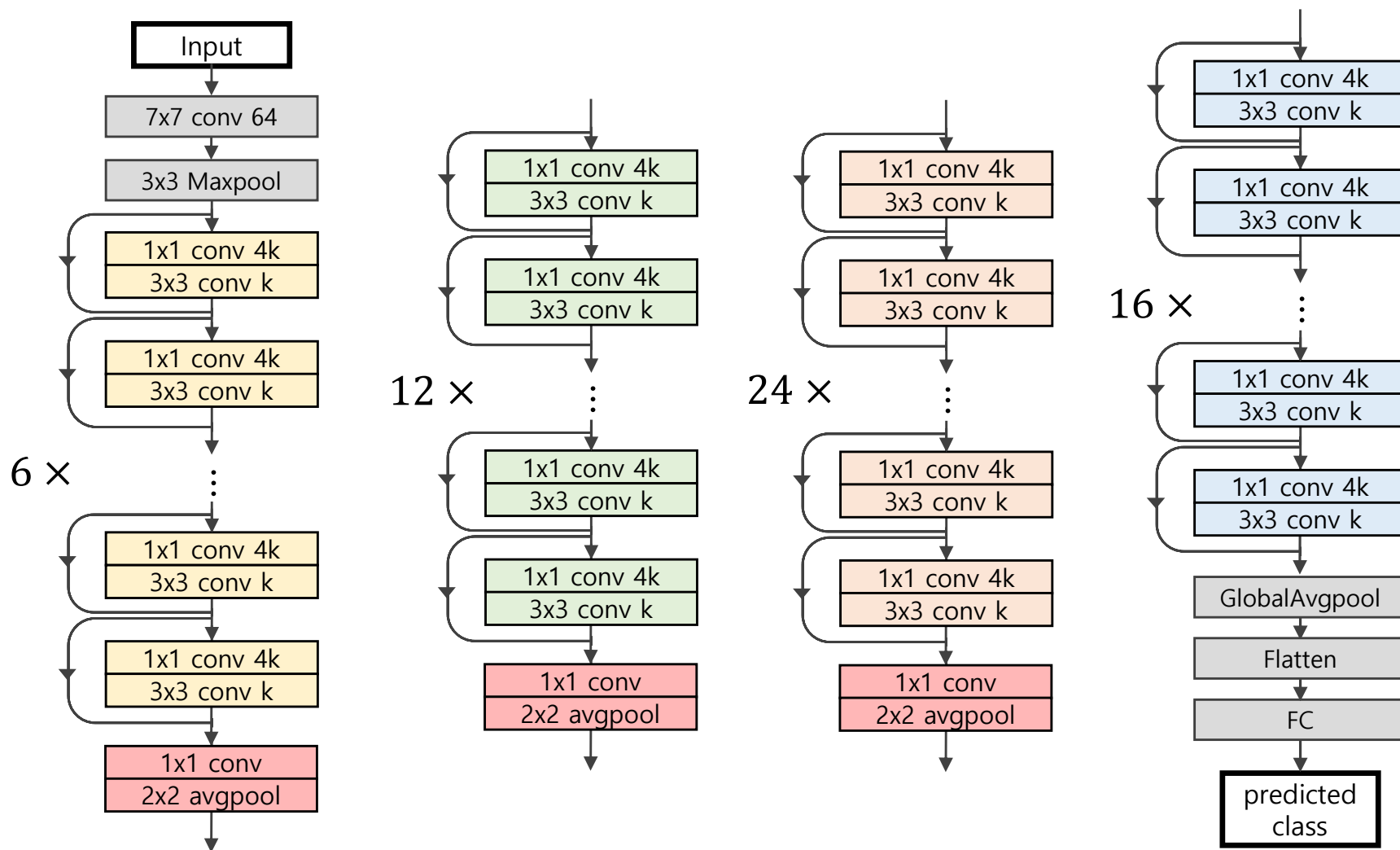


Concatenation is possible only when the number of channels is equal
⇒ Channels can't be regulated in a dense block
⇒ The number of channels is halved in the transition layer

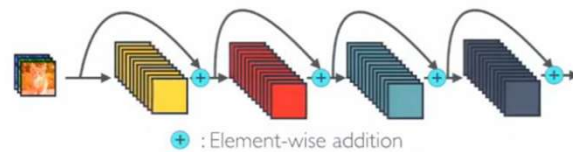


Architecture

3. Architecture of DenseNet121

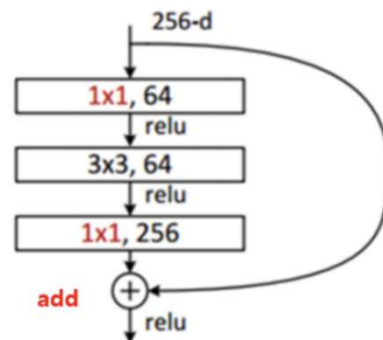


ResNet vs. DenseNet

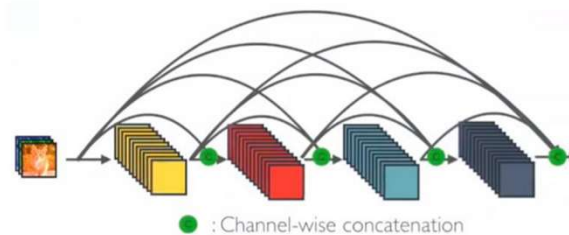


**Skip Connection
Addition**

$$x_l = H_l(x_{l-1}) + x_{l-1}$$

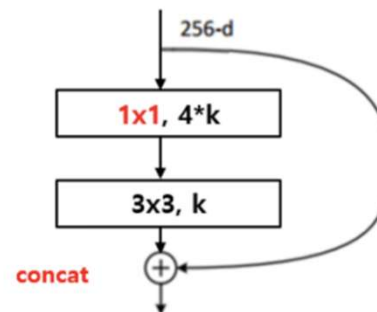


**bottleneck
(for ResNet)**



**Dense Connection
Concatenating**

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}])$$



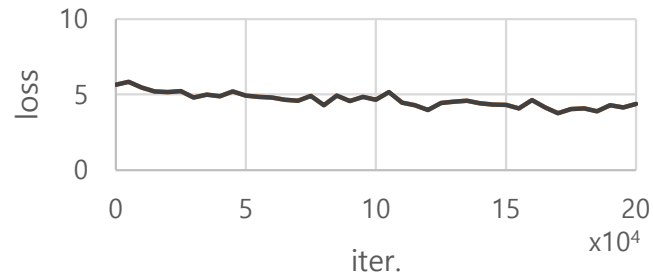
**bottleneck
(for DenseNet)**

■ Setup

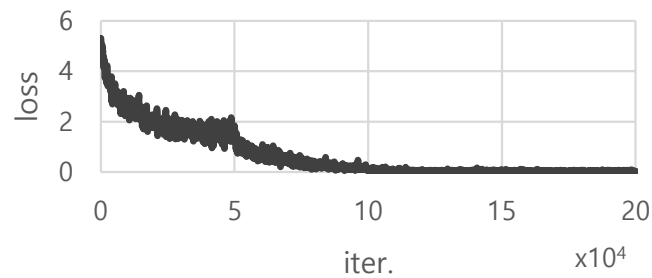
	ResNet50	DenseNet121, 201
Dataset	Tiny ImageNet	
Optimizer	SGD	
Learning rate	~ 50,000 ~ 70,000 ~ 100,000 ~ 125,000 ~ 150,000 ~ 175,000 ~ 200,000	0.1 0.01 0.005 0.001 0.0001 0.00005 0.00001
Batch Size	64	
Iteration	200,000	
Augmentation	Horizontal flip Rotate 90 clockwise	
Hardware	NVIDIA GTX 2080 Ti GPU	

■ Result

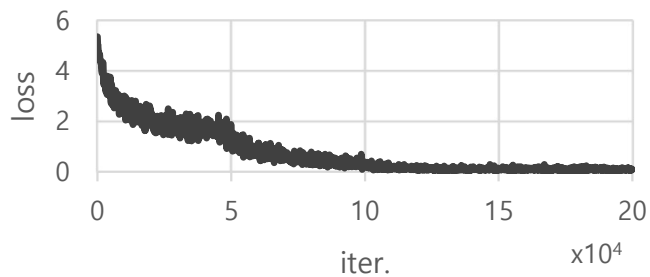
ResNet50 Loss



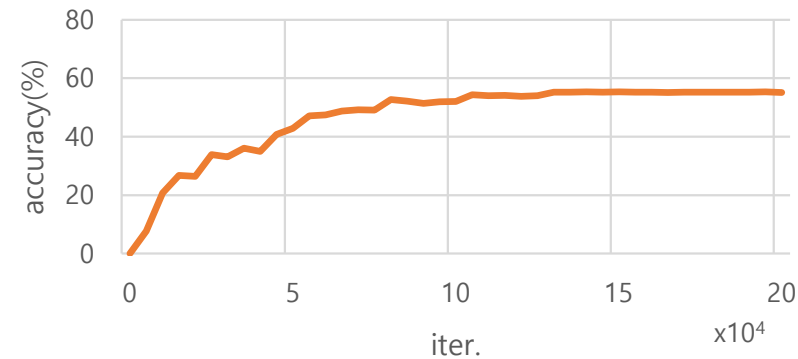
DenseNet201



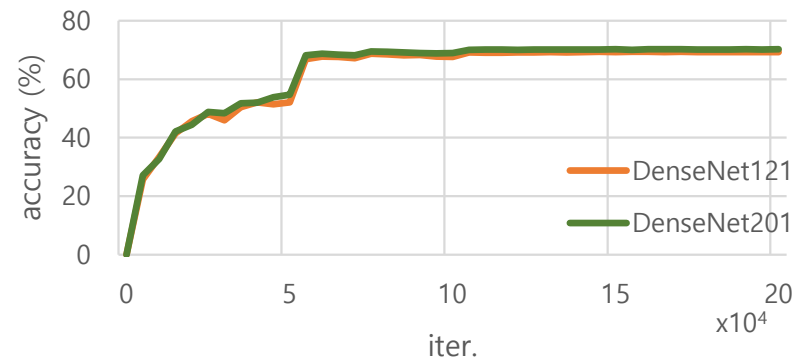
DenseNet121



Accuracy



DenseNet Accuracy



	Accuracy (%)	Loss
ResNet50	55.30	0.1155
DenseNet121	69.45	0.0216
DenseNet201	70.30	0.0086

Future work